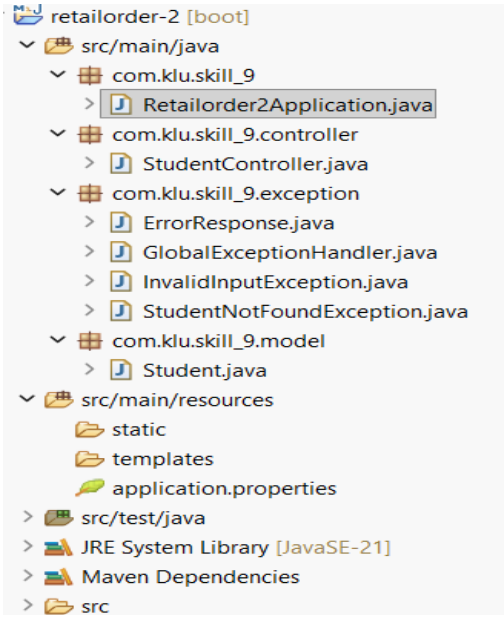
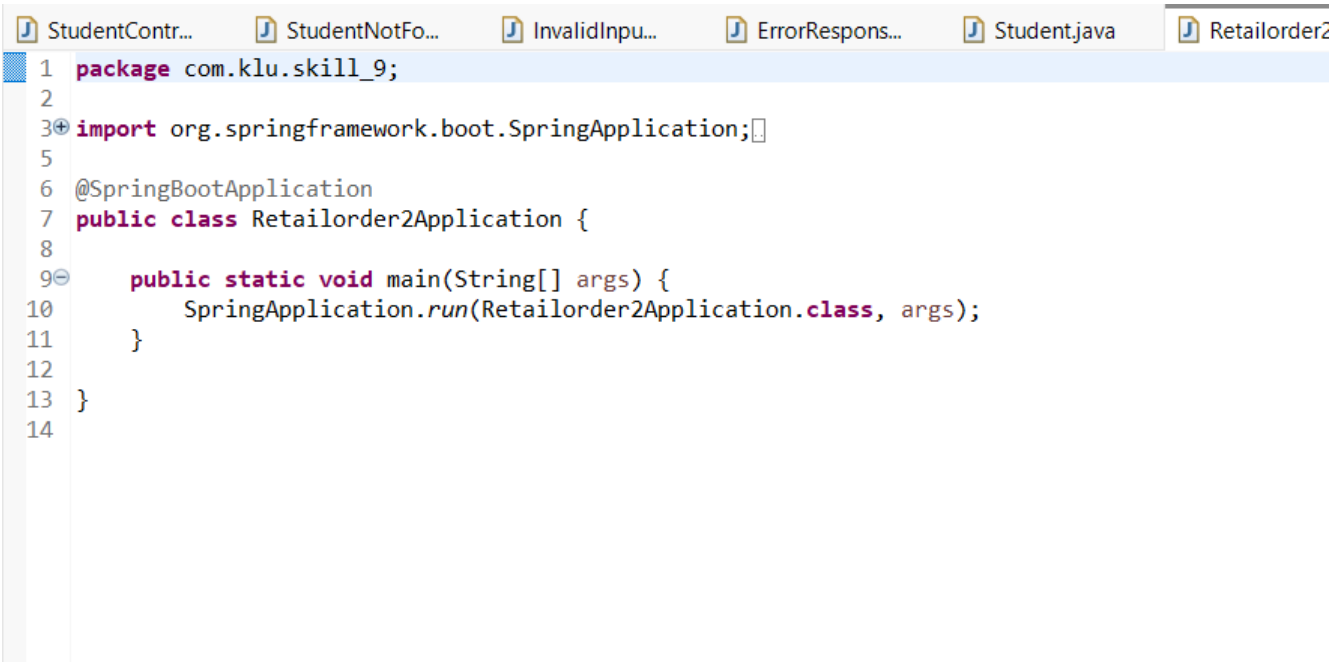
NAME:HUMAIRA TASNEEM
ID NO.2400030278
SECTION NO:4

SKILL - 9 Global Exception Handling using @ControllerAdvice

Project structure :



APPLICATION.JAVA:




```java
package com.klu.skill_9;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class Retailorder2Application {

    public static void main(String[] args) {
        SpringApplication.run(Retailorder2Application.class, args);
    }

}
```

STUDENT CONTROLLER.JAVA:

```java
package com.klu.skill_9.controller;

import com.klu.skill_9.model.Student;
import com.klu.skill_9.exception.InvalidInputException;
import com.klu.skill_9.exception.StudentNotFoundException;

import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/api")
public class StudentController {

    @GetMapping("/student/{id}")
    public Student getStudent(@PathVariable String id) {

        // Step 1: Check numeric input
        if (!id.matches("\\d+")) {
            throw new InvalidInputException("Invalid input! ID must be a number.");
        }

        int studentId = Integer.parseInt(id);

        if (studentId != 1) {
            throw new StudentNotFoundException("Student not found with ID: " + studentId);
        }
        return new Student(1, "Maira");
    }
}
```

ERROR RESPONSE.JAVA:

```java
package com.klu.skill_9.exception;

import java.time.LocalDateTime;

public class ErrorResponse {

    private LocalDateTime timestamp;
    private String message;
    private int statusCode;

    public ErrorResponse(LocalDateTime timestamp, String message, int statusCode) {
        this.timestamp = timestamp;
        this.message = message;
        this.statusCode = statusCode;
    }

    public LocalDateTime getTimestamp() {
        return timestamp;
    }

    public String getMessage() {
        return message;
    }

    public int getStatusCode() {
        return statusCode;
    }
}
```

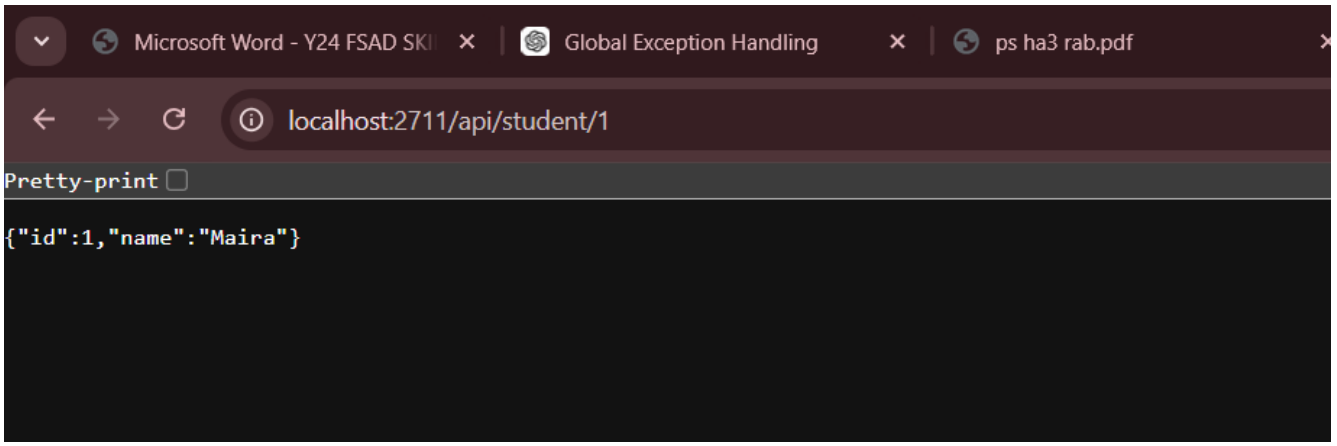
STUDENTNOTFOUNDEXCEPTION.JAVA:

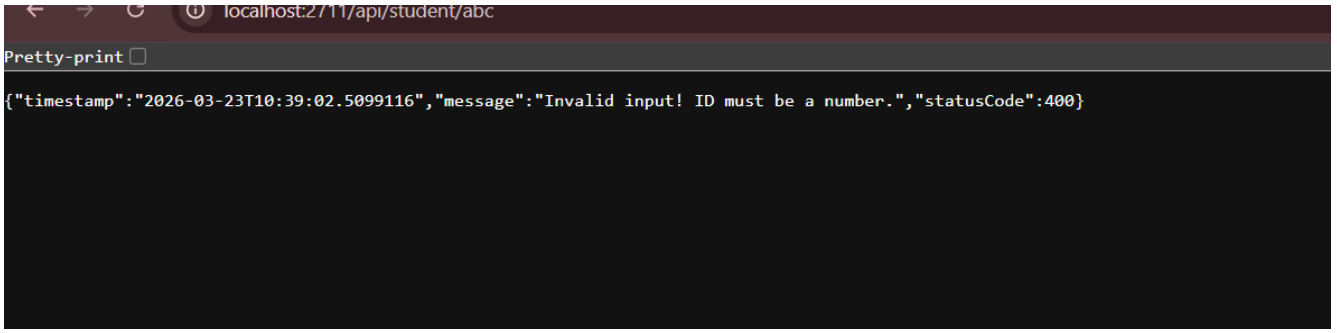
```java
package com.klu.skill_9.exception;

public class StudentNotFoundException extends RuntimeException {

    /**
     *
     */
    private static final long serialVersionUID = 1L;

    public StudentNotFoundException(String message) {
        super(message);
    }
}
```

STUDENT.JAVA:

```java
package com.klu.skill_9.model;

public class Student {

    private int id;
    private String name;

    // Constructor
    public Student(int id, String name) {
        this.id = id;
        this.name = name;
    }

    // Getter for ID
    public int getId() {
        return id;
    }

    // Getter for Name
    public String getName() {
        return name;
    }
}
```

OUTPUT:





Repo link: https://github.com/Tasneem1027/FSAD_SkillLab_6-11